

Question ID: 25faa756

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number a is 60% greater than the positive number b . The number c is 45% less than a . The number c is how many times b ?

Correct Answer: .88, 22/25

Rationale

The correct answer is .88. It's given that the number a is 60% greater than the positive number b . Therefore, $a = (1 + \frac{60}{100})b$, which is equivalent to $a = (1 + 0.60)b$, or $a = 1.60b$. It's also given that the number c is 45% less than a . Therefore, $c = (1 - \frac{45}{100})a$, which is equivalent to $c = (1 - 0.45)a$, or $c = 0.55a$. Since $a = 1.60b$, substituting $1.60b$ for a in the equation $c = 0.55a$ yields $c = 0.55(1.60b)$, or $c = 0.88b$. Thus, the number c is 0.88 times the number b . Note that .88 and 22/25 are examples of ways to enter a correct answer.